

Lab-10

Study of IP Addressing and sub-netting.

1. Find default subnet masks, network bits, host bits, hosts per subnet, no of subnets, subnet number, 1st valid IP address, last valid IP address, and broadcast address.

i) 8.1.4.5/16

Class: A

Default mask: 255.0.0.0

Given mask: /16 $\Rightarrow$ 255.255.0.0

Network bits: 16

Host bits: 16

Host per subnet: $2^{16} - 2 = 65534$ Number of subnets: borrowed bits = $16 - 8 = 8 \Rightarrow 2^8 = 256$ subnets

Subnet address: 8.1.0.0

1st valid host: 8.1.0.1

Last valid host: 8.1.255.254

Broadcast address: 8.1.255.255

ii) 130.4.102.1/24

Class: B

Default mask: 255.255.0.0

Given mask: /24 $\Rightarrow$ 255.255.255.0

Network bits = 24, Host bits = 8

Hosts per subnets: borrowed = $24 - 16 = 8 \Rightarrow 2^8 = 256$ subnets

Subnet address: 130.4.102.0

1st valid host: 130.4.102.1

Last valid host: 130.4.102.254

Broadcast: 130.4.102.255

iii) 130.4.102.1/22

Class: B

Default mask: 255.255.0.0

Given mask: /22 $\Rightarrow$ 255.255.255.0

Network bits: 22

Host bits: 10

Host per subnet: $2^{10} - 2 = 1022$

Number of subnets: borrowed = $22 - 16 = 6 \Rightarrow 2^6 = 64$

Subnet block size in 3rd octet: $256 - 252 = 4$

Network address: 130.4.100.0

1st valid: 130.4.100.1

Last valid: 130.4.102.254

Broadcast: 130.4.103.255

iv) 199.1.1.100/27

Class: C

Default mask: 255.255.255.0

Given mask: /27 $\Rightarrow$ 255.255.255.224

Network bits: 27, Host bits: 5

Hosts per subnet: $2^5 - 2 = 30$

Number of subnets: borrowed = $27 - 24 = 3 \Rightarrow 2^3 = 8$ subnets

Network address: 199.1.1.96

1st valid: 199.1.1.97

Last valid: 199.1.1.126
Broadcast: 199.1.1.127

2. A host in a class C network has been assigned an IP address 192.168.17.9. Find the number of addresses in the block, the first address and the last address.

Ans. 192.168.17.9

Class: C

Default mask: 255.255.255.0

Number of addresses in block: $2^8 = 256$, usable hosts 254

First address: 192.168.17.0

Last address: 192.168.17.255

3. An address in a block is given as 185.28.17.9. Find the number of addresses in the block, the first address and the last address.

Ans. 185.28.17.9

Default mask: 255.255.0.0

Number of addresses in block: $2^{16} = 65536$, usable hosts 65534

First address: 185.28.0.0

Last address: 185.28.255.255

4. A block of addresses is granted to a small organization. We know that one of the addresses is 206.16.37.39/28. What is the first address, last address, number of addresses in a block?

Ans. Given 205.16.37.39/28
 Mask: /28 $\Rightarrow$ 255.255.255.240
 Addresses per block: $2^{(32-28)} = 2^4 = 16$
 First valid host: 205.16.37.33
 Last valid host: 205.16.37.46
 Broadcast: 205.16.37.47

5. Subnet the IP address 216.21.5.0 into 30 hosts in each subnet. Find class, Default Mask, subnet mask, Bit borrowed, new subnet mask, no. of hosts & Subnet, network ranges.

Ans. Given 216.21.5.0 into 30 hosts per subnet.

Class: C

Default Mask: 255.255.255.0 (/24)

Required Hosts: 30

Hosts needed: $30 + 2 = 32$

Host Bits Required: 5

Subnet mask: $32 - 5 = /27 = 255.255.255.224$

Bit borrowed: 3

No. of subnets: $2^3 = 8$

Hosts per Subnet: $2^5 - 2 = 30$

Subnet Ranges:-

1) 216.21.5.0 - 216.21.5.31

2) 216.21.5.32 - 216.21.5.63

3) 216.21.5.64 - 216.21.5.95

4) 216.21.5.96 - 216.21.5.127

5) 216.21.5.128 - 216.21.5.159

6) 216.21.5.160 - 216.21.5.191

7) 216.21.5.192 - 216.21.5.223

8) 216.21.5.224 - 216.21.5.255

6. Subnet the IP address 192.10.20.0 into 52 hosts in each subnet. Find class, Default Mask, Bit borrowed, new subnet mask, no. of hosts & subnet, network ranges (Subnets).

Ans. Class: C

Default mask: 255.255.255.0 (124)

Required hosts: 52

Hosts needed: $52 + 2 = 54$

Host bits required: 6

Subnet mask: $32 - 6 = 26 = 255.255.255.192$

Bits borrowed: 2

No. of subnets: $2^2 = 4$

Hosts per Subnet: $2^6 - 2 = 62$

Subnet Ranges:

1) 192.10.20.0 - 192.10.20.63

2) 192.10.20.64 - 192.10.20.127

3) 192.10.20.128 - 192.10.20.191

4) 192.10.20.192 - 192.10.20.255

7. Determining the subnet mask for Devices A and B:

a) Device A: 172.16.17.30/20

Class: B

Default Mask: 255.255.0.0 (/16)

Given Mask: /20 = 255.255.240.0

Block size: 16 (256 - 240)

Network addresses: 172.16.16.0

First valid host: 172.16.16.1

Last valid host: 172.16.31.254

Broadcast: 172.16.31.255

b) Device B: 172.16.28.15/20

Class: B

Default Mask: 255.255.0.0 (/16)

Given Mask: /20 = 255.255.240.0

Block size: 16

Network addresses: 172.16.16.0

First valid host: 172.16.16.1

Last valid host: 172.16.31.254

Broadcast: 172.16.31.255

Device A and B belong to the same subnet 172.16.16.0/20.